

Uniform Random Variables with Uniform Sum

GREEN-071

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Suppose X, Y are finitely-supported independent random variables taking integer values such that $X + Y$ is uniformly distributed on its range. Are X and Y themselves uniformly distributed on their ranges?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.
Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>